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Animal-care telemetry and human review
Technical architecture discussion
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Sanctuary operating context

Veterinary imaging, rehabilitation and nutrition create different records.

Public descriptions establish care capabilities. They do not establish internal technology gaps.

Discovery needs an institutional data owner and an approved workflow.

The integration question

Which handoff loses identity, time or source context?

A bounded review layer can sit alongside existing clinical systems.

Sensor selection depends on species, welfare, placement and device capability.

Six synthetic metadata adapters

Camera events and imaging metadata
Audio metadata and device telemetry
Laboratory results and care tasks

Working input: JSON fixtures. Live media and protocol connectors remain proposed.

Deterministic record validation

The importer checks schema, source provenance, animal identity and timestamps.

Duplicate events remain idempotent. Conflicting content triggers rejection.

Atomic batches prevent partial imports. Species-specific physiological baselines remain proposed.

Accountable review workflow

A source event enters validation, then persistent storage.

A rule may create a review task. An operator can acknowledge that task with a source-linked audit entry.

Acknowledgment records review activity. A veterinarian retains clinical responsibility.

Vision and gait research

Proposed input: approved video with species, camera calibration and expert labels.

Evaluation needs held-out animals and sites, occlusion tests and measurement repeatability.

Pose landmarks, thermal interpretation and gait symmetry require separate validation.

Acoustic research requirements

Proposed input: consented recordings, device frequency response and expert annotation.

A spectrogram can describe signal energy. A pain inference requires independent welfare validation.

Infrasound capture depends on suitable hardware. Ordinary microphones may not capture it.

Cross-department handoffs

Current example: a source laboratory flag opens a review task.

Proposed extension: an authorized reviewer assigns a follow-up to the relevant care team.

Nutrition changes require a qualified author and the institution’s approved workflow.

Deployment and data boundaries

Local operator: Docker, Python and SQLite, with persistent runtime storage.

Public website: a separate static synthetic snapshot on Cloudflare Pages, planned under /bio/.

Production needs named users, access controls, tested recovery and an agreed hosting boundary.

Synthetic verification evidence

16 core tests and 7 HTTP checks passed in the recorded local run.

The snapshot contains 50 fictional animals, 359 events and 5 open review tasks.

Public route checks covered read-only access and rejection of operator routes.

A focused engineering walkthrough

20 minutes with a veterinary lead and an IT/data owner.

Review one workflow, one permitted sample export and one accountable reviewer. Agree an import-fidelity measure before a pilot.

Next gate: written scope, data rights, security review and a supervised read-only evaluation.